Diphtheria, tetanus and pertussis: GBD 2023 estimates vs WHO reported cases

How much does the implied reporting rate vary between countries and over time?

22 October 2026

We compare GBD 2023 modelled incidence with case counts reported to WHO for diphtheria, tetanus and pertussis in 204 countries, 1990-2023. The ratio of the two, the *implied reporting rate*, varies widely both between and within countries.

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1 Summary

- **Reported cases are a small and unstable fraction of modelled cases.** The median country-year reporting rate is 0.6% for pertussis, 14% for diphtheria and 19% for tetanus.

- **The rate moves a lot within countries.** In a typical country, the highest year's rate is 134x the lowest for pertussis, 42x for diphtheria and 20x for tetanus. Change over time within countries accounts for 36%-51% of all variation in the (log) rate; the rest is stable differences between countries.
- **Year-to-year changes in reported cases barely track GBD's changes.** The correlation of annual log changes is 0.2 for diphtheria, 0.16 for pertussis and 0.02 for tetanus. GBD's estimates are smooth over time, so year-to-year swings in the rate come almost entirely from reported cases.
- **DTP3 coverage explains differences between countries more than changes within them.** Countries with higher coverage report a larger share of modelled cases (Spearman 0.32-0.42), but within a country, coverage and reporting rate move together only weakly.
- **The ratio means something different for each disease** (Section 8). For pertussis, GBD scales WHO notifications up to Switzerland's reporting level, so the ratio mostly measures distance from Switzerland (GBD 2023 Disease and Injury and Risk Factor Collaborators 2025). For diphtheria, GBD 2023 replaces its estimate with the WHO count wherever the model falls below it, which is why 144 diphtheria country-years match almost exactly. For diphtheria and tetanus, GBD derives incidence from deaths divided by a case-fatality ratio, so the ratio compares notifications with a death-based estimate.
- **Reports above GBD's estimate are mostly a "GBD expects almost nothing" effect.** In 2019-2023, 72% of diphtheria and 70% of tetanus country-years that report more than GBD are places where GBD estimates fewer than 10 cases, mostly in Europe and the Americas. The exceptions with a real GBD burden are outbreak settings (diphtheria in Nigeria, Niger and India in 2023) and tetanus in East and Central Africa.
- **Large "silent" gaps are mostly missing reports, not zeros.** Of the country-years where GBD estimates at least 100 cases a year and WHO has no positive count, 60% are blanks rather than reported zeros. The largest is pertussis in Nigeria: no report to WHO in any year 2019-2023, against about 1,920,000 modelled cases a year.

2 Data and methods

2.1 Sources

Table 1: Data sources.

Source	Measures	Years	Breakdown
GBD 2023 Results Tool	Incidence and deaths (number, rate) with 95% UI	1990-2023	Country; all ages, both sexes
WHO GHO	Reported cases; DTP1/DTP3 coverage (WUENIC)	1974-2025	Country-year only

GBD estimates come from GBD 2023 Results Tool exports for 204 countries. Incidence is available annually for 1990-2023. GHO case counts are downloaded from the GHO OData API: diphtheria (`WHS3_41`), pertussis (`WHS3_43`) and total tetanus (`WHS3_46`, which includes neonatal tetanus and is compared with GBD "Tetanus"). GHO does not break these counts down by age or sex, so all comparisons use all-age, both-sex totals. Countries are matched on ISO3 codes. 14 territories appear only in GHO and 4 only in GBD (Appendix, Table 10).

2.2 Implied reporting rate

For each country, cause and year, the **implied reporting rate** is GHO reported cases divided by GBD modelled incidence. It is computed only where GBD estimates at least 10 cases, because below that the ratio is mostly noise. Table 2 shows how many country-years each rule leaves.

Table 2: Country-years 1990-2023 by whether a reporting rate can be computed.

Cause	Rate computed	Zero reported	GBD < 10 cases	Not reported to WHO	Reporting break
Diphtheria	1,115	901	3,442	1,478	0
Pertussis	3,938	1,498	71	1,429	0
Tetanus	2,288	248	2,849	1,547	4

Rates are analysed on the $\log_{10}$ scale, so a move from 1% to 10% counts the same as a move from 10% to 100%. Years in which a country reported zero cases have no log rate. They are shown separately and are not imputed. Per-country variability is summarised for countries with at least four years with a rate, using:

- the **fold range** (highest ÷ lowest non-zero rate);
- the SD of $\log_{10}$ rate;
- a linear **trend** in $\log_{10}$ rate per decade.

The **within-country share** is the proportion of the total sum of squares of $\log_{10}$ rate that lies around each country's own mean.

2.3 Data-quality flags

Following the research brief (VIMC GBD–WHO comparison project 2026), three kinds of country-year are flagged:

- **Known reporting breaks.** India's "total tetanus" loses its non-neonatal cases from 2020 and equals the neonatal count from 2022, so India tetanus 2020 onwards is excluded from all rates. Pakistan pertussis 1989-1990 and China pertussis from 2017 (PCR accepted) are labelled but kept.
- **GBD floored at the WHO count.** GBD 2023 replaces its diphtheria and pertussis estimates with the WHO count wherever the model falls below it (GBD 2023 Disease and Injury and Risk Factor Collaborators 2025). Country-years within 1% of the WHO count are labelled as floored: 144 for diphtheria and 48 for pertussis. They stay in the rate analysis but are shown separately in the country-level breakdowns.
- **Missing vs zero.** GHO stores a missing report as null and a reported zero as 0. The pipeline keeps the two apart throughout.

2.4 Interactive explorer

`explorer.html` in the repository is an interactive companion to this report. It shows Figure 1 for any year with every country named on hover, breaks Figure 19 down by country, and gives a profile for each country with both sources over time. Open it in a web browser.

3 Reported vs modelled cases

Figure 1 compares the two sources country by country. Dashed lines mark reported = 100%, 10% and 1% of modelled, from left to right. Pertussis reports sit one to three orders of magnitude below GBD's estimates. Diphtheria and tetanus scatter around the 10-100% lines, and many diphtheria points lie exactly on the 100% line. Many countries report zero cases even where GBD estimates hundreds or thousands.

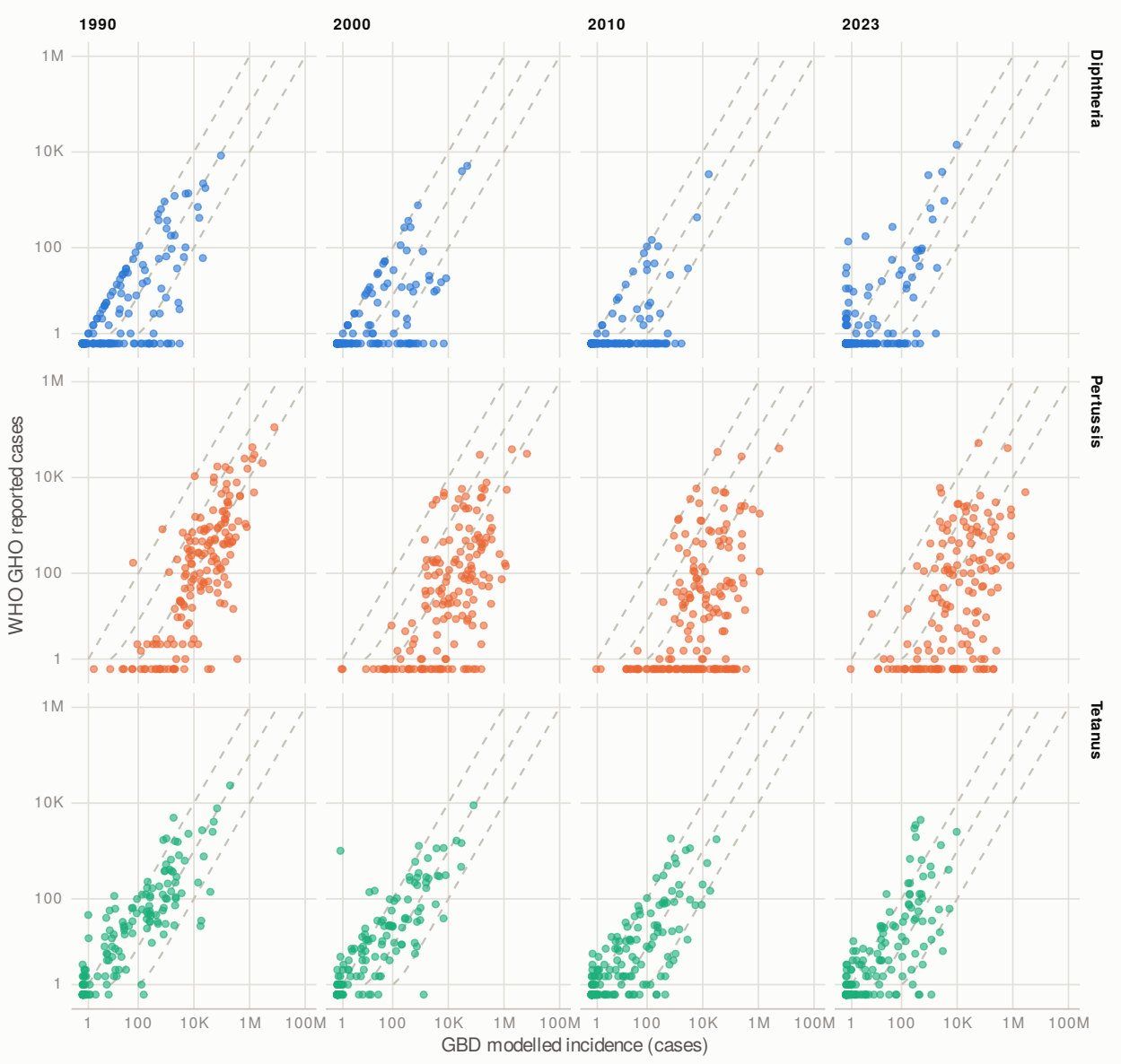


Figure 1: WHO-reported cases vs GBD modelled incidence by country, selected years. Dashed lines, left to right: reported = 100%, 10% and 1% of modelled. Points on the bottom edge reported zero cases.

3.1 Which countries sit where

To see which countries make up each part of Figure 1, every country-year is put into one of seven classes by how the WHO count compares with GBD's estimate: not reported to WHO, zero reported, below 1%, 1-10%, 10-99%, equal to GBD because of the floor, and above GBD. This classification keeps country-years with fewer than 10 modelled cases, so every point has a class. Figure 2 and Figure 3 show the mix for 2019-2023.

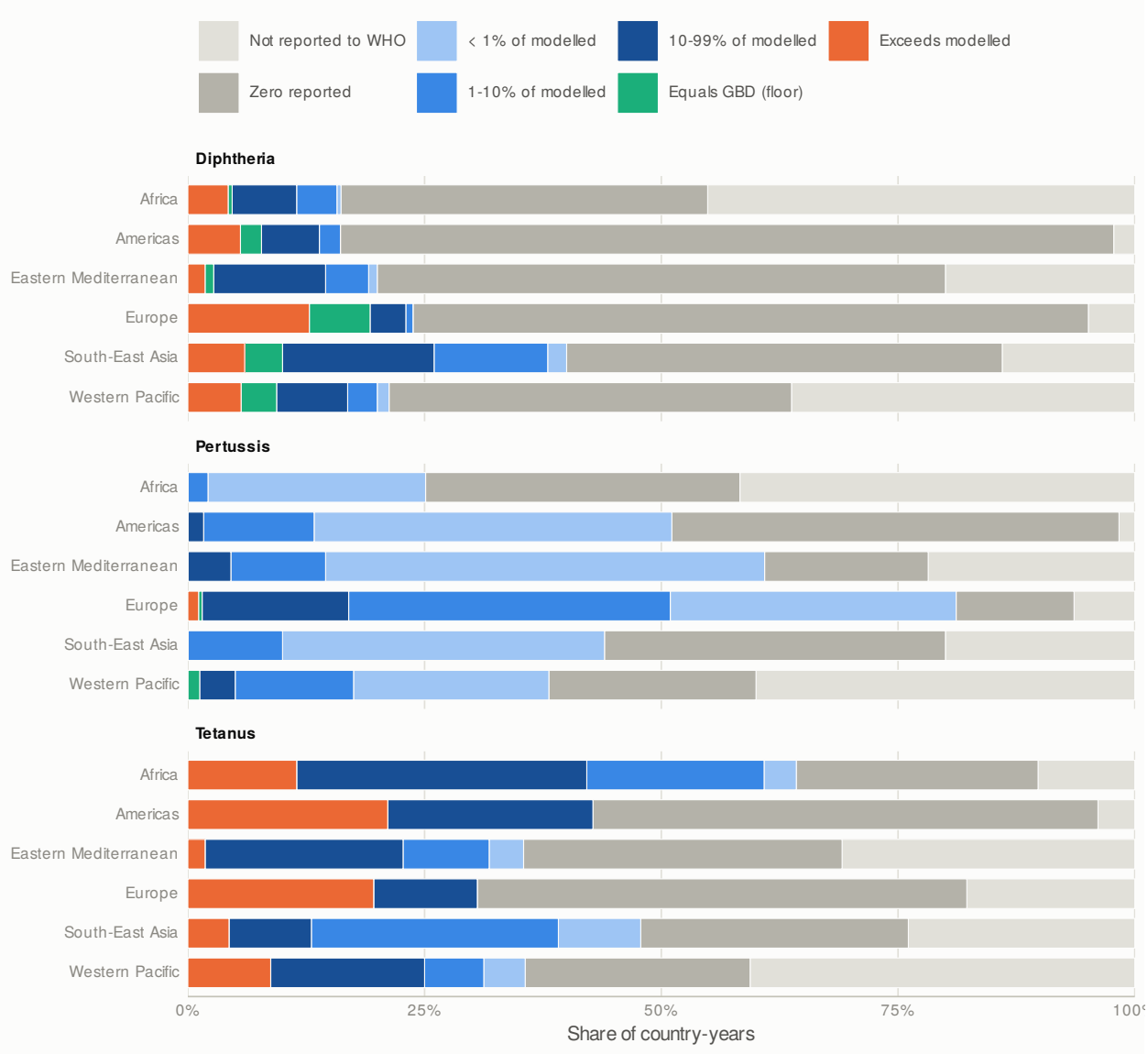


Figure 2: How WHO counts compare with GBD estimates, by WHO region, 2019-2023 country-years.

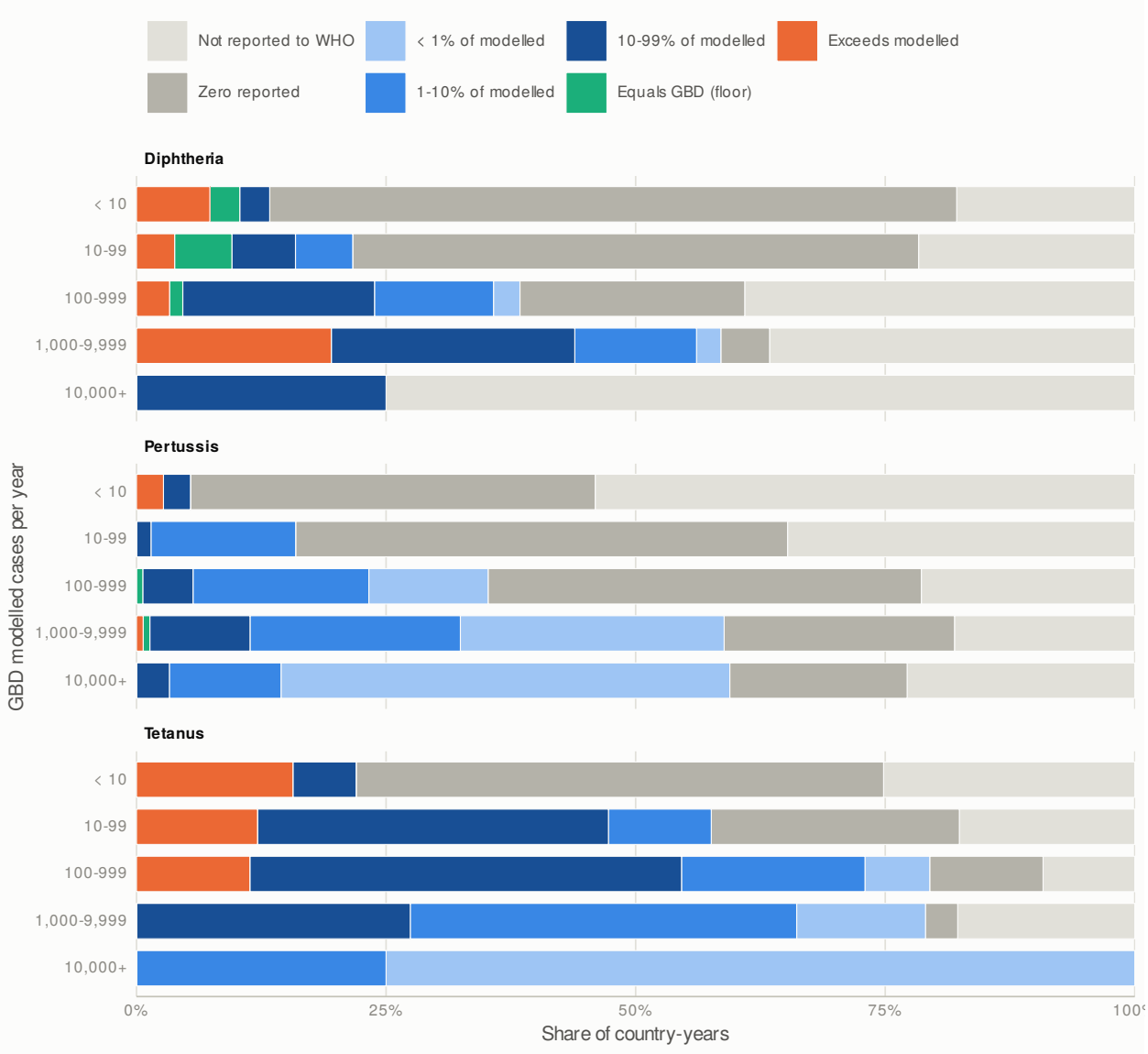


Figure 3: How WHO counts compare with GBD estimates, by size of GBD's estimate, 2019-2023 country-years.

Four patterns stand out.

1. **Zero reports cluster where GBD expects few cases.** For diphtheria, 67% of country-years where GBD estimates fewer than 100 cases report zero, against 4% where it estimates 1,000 or more; for tetanus the figures are 47% and 3%. Pertussis is the exception: 20% of country-years report zero even where GBD estimates 1,000 cases or more.
2. **Reports above GBD come from two different places.** Most are in Europe and the Americas, where GBD estimates almost no cases: 33 of the 34 European diphtheria country-years above GBD, and 52 of 52 for tetanus, have GBD estimates below 10 cases. The rest are outbreaks or high-burden settings, chiefly diphtheria in Nigeria, Niger and India and tetanus in Uganda, Kenya, Angola and the Central African Republic.
3. **Pertussis almost never matches.** Across all regions and burden levels, most pertussis reports are below 10% of GBD's estimate. Only Europe has a substantial share in the 10-99% band.
4. **The largest gaps are missing reports.** Table 3 lists countries where GBD estimates at least 100 cases a year but WHO received no positive count in 2019-2023. Most of these country-years are blanks rather than reported zeros.

Table 3: Countries with no positive WHO count in 2019-2023 despite a GBD estimate of at least 100 cases a year (top 8 per disease by GBD estimate). Zero = years reported as zero; Missing = years with no report.

Disease	Country	WHO region	GBD cases/year	Zero	Missing	DTP3 (%)
Diphtheria	Angola	Africa	1,324	0	5	55
	Mali	Africa	496	3	2	75
	Cameroon	Africa	492	0	5	70
	United Republic of Tanzania	Africa	206	0	5	88
	Benin	Africa	197	1	4	70
	Mozambique	Africa	190	1	4	67
	Burundi	Africa	166	0	5	92
	Malawi	Africa	159	2	3	92
Pertussis	Nigeria	Africa	1,915,462	0	5	59
	Somalia	Eastern Mediterranean	303,405	2	3	63
	Angola	Africa	295,985	2	3	55
	Chad	Africa	209,869	0	5	61
	United Republic of Tanzania	Africa	197,877	0	5	88
	Cameroon	Africa	189,905	0	5	70
	Egypt	Eastern Mediterranean	117,784	5	0	96
	Mozambique	Africa	71,701	1	4	67
Tetanus	Somalia	Eastern Mediterranean	3,787	0	5	63
	Democratic People's Republic of Korea	South-East Asia	185	1	4	98

3.2 The latest year in detail

Figure 4, Figure 5 and Figure 6 repeat the 2023 panels of Figure 1 with notable countries named. The labels mark every country that reported more than GBD estimates, plus the largest GBD burdens among countries reporting zero or far below.

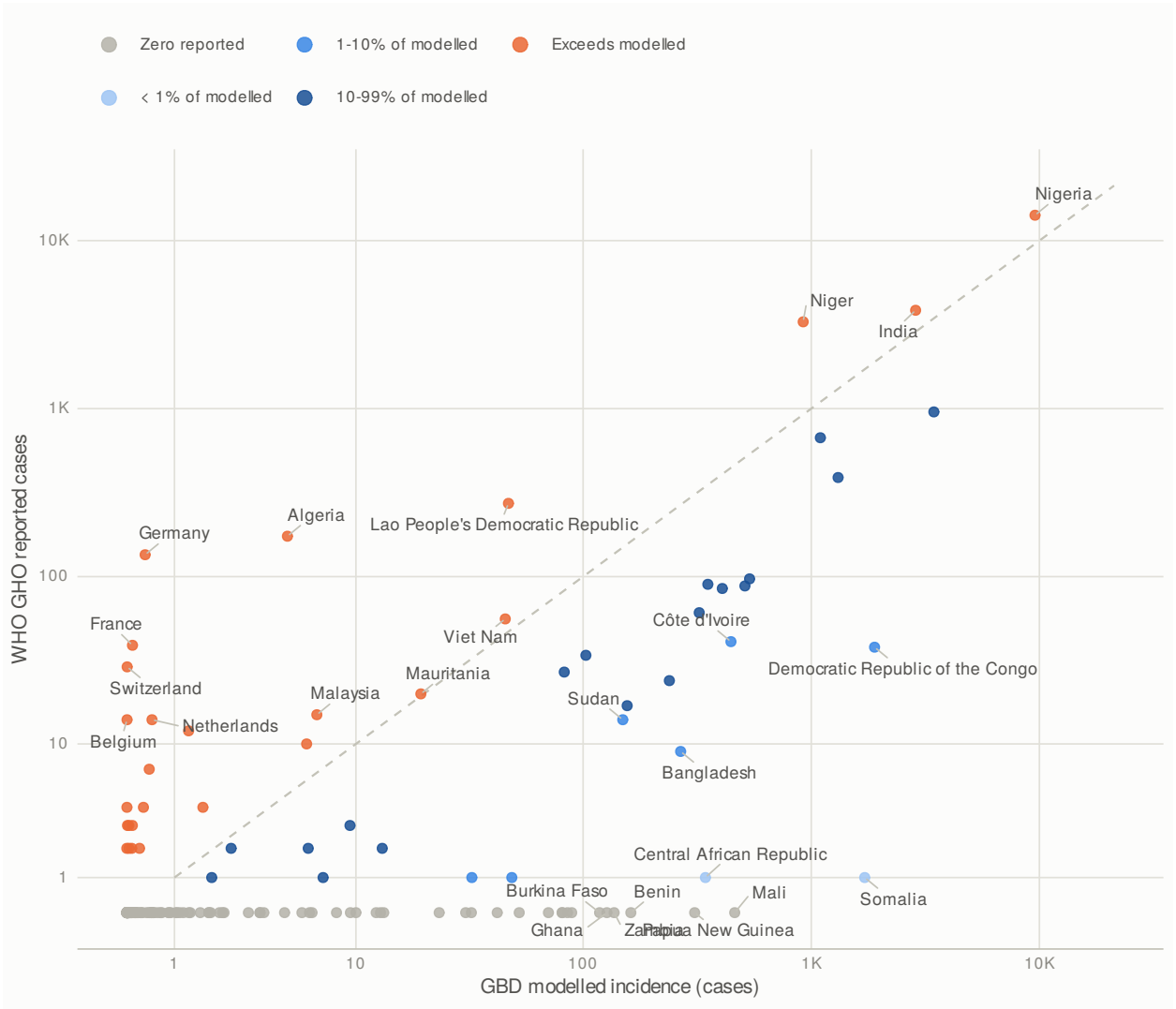


Figure 4: Diphtheria, 2023: WHO-reported vs GBD-modelled cases with notable countries labelled. Dashed line: reported = modelled. Points on the bottom row reported zero.

For diphtheria in 2023, the cluster at the top left is mostly Western Europe. Germany, France, Switzerland, Belgium and the Netherlands each reported between 14 and 135 cases while GBD estimates less than one. GBD derives diphtheria incidence from deaths, so with almost no deaths it estimates almost no cases. Switzerland's recent cases were mostly cutaneous infections and carriers among asylum seekers (Brockhaus et al. 2024), and GBD's definition is respiratory diphtheria; a similar case mix elsewhere in Western Europe is plausible but not verified here. Nigeria, Niger and India report more than GBD because of documented outbreaks (Bita Fouda et al. 2025). At the bottom right, Mali, Papua New Guinea, Benin, Zambia, Ghana and Burkina Faso reported zero against GBD estimates of 118-462 cases.

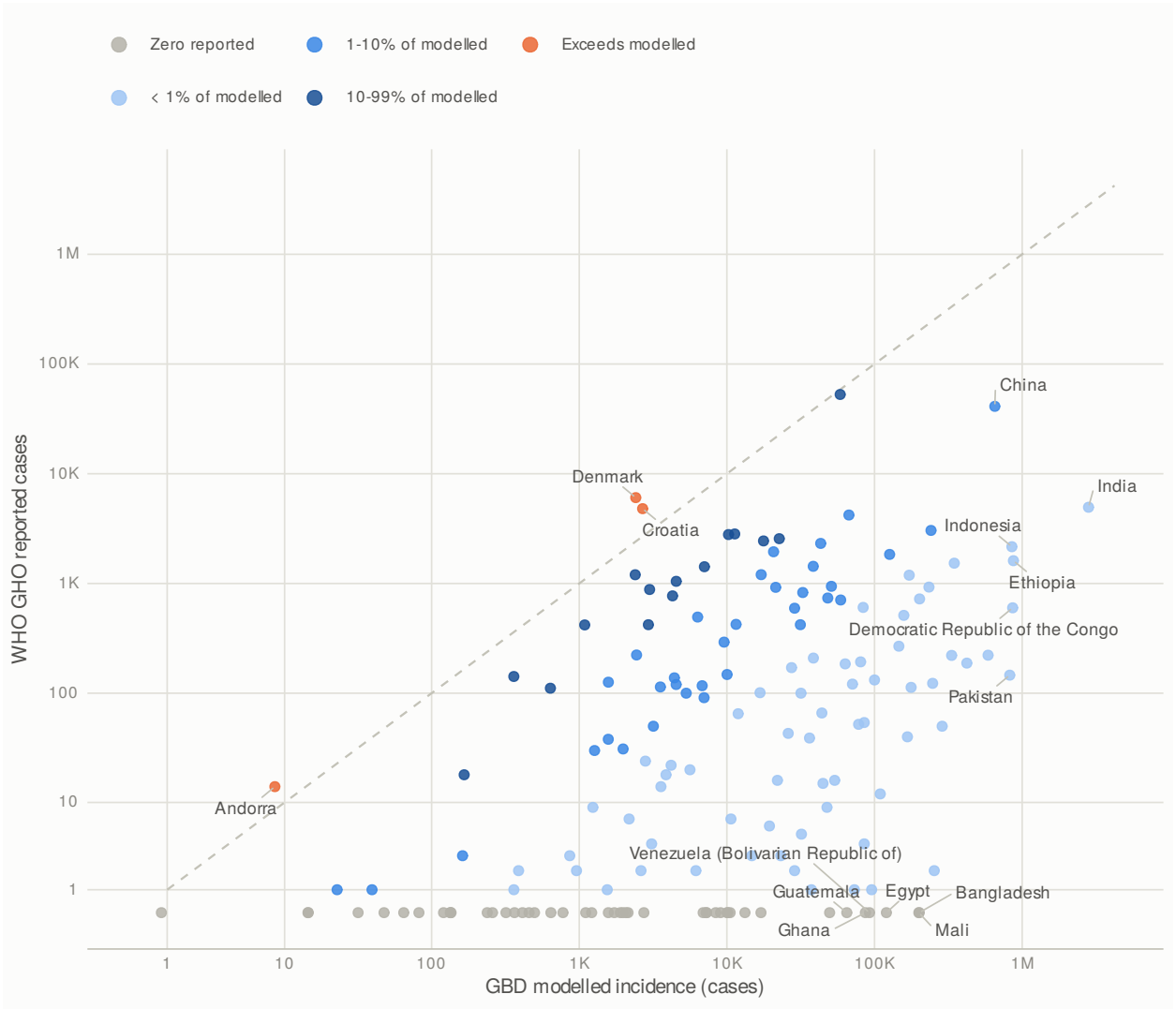


Figure 5: Pertussis, 2023: WHO-reported vs GBD-modelled cases with notable countries labelled.

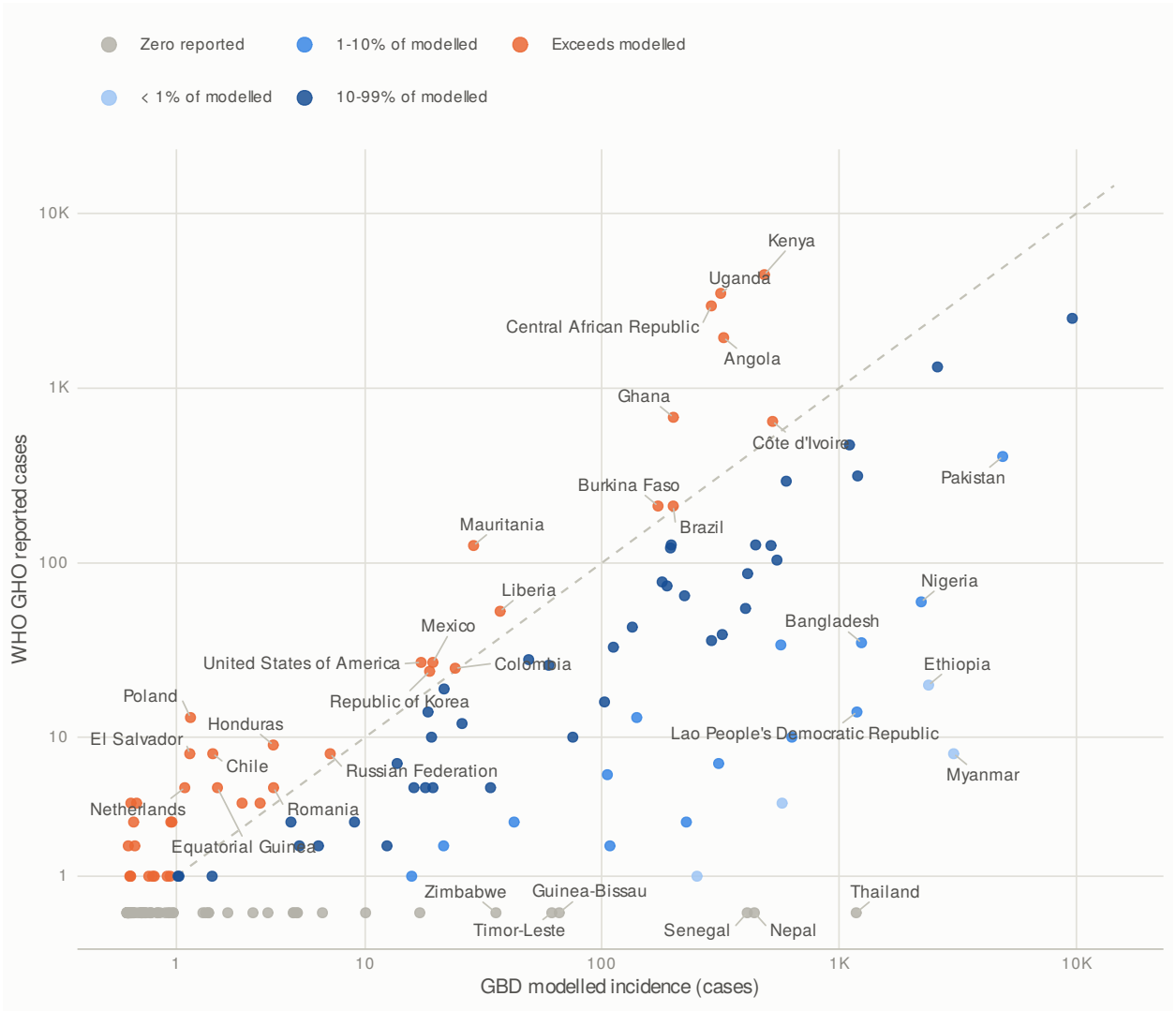


Figure 6: Tetanus, 2023: WHO-reported vs GBD-modelled cases with notable countries labelled.

Table 4: Country-years in 2023 where WHO reported more cases than GBD estimates (excluding the GBD floor), top 8 per disease by reported cases.

Disease	Country	WHO region	WHO reported	GBD modelled	Ratio
Diphtheria	Nigeria	Africa	14,163	9,565	1.5x
	India	South-East Asia	3,850	2,867	1.3x
	Niger	Africa	3,286	922	3.6x
	Lao People's Democratic Republic	Western Pacific	273	47	5.8x
	Algeria	Africa	174	4.9	36x
	Germany	Europe	135	0.4	>100x
	Viet Nam	Western Pacific	56	46	1.2x
	France	Europe	39	0.1	>100x
Pertussis	Denmark	Europe	6,059	2,414	2.5x
	Croatia	Europe	4,806	2,686	1.8x
	Andorra	Europe	14	8.6	1.6x
Tetanus	Kenya	Africa	4,463	485	9.2x
	Uganda	Africa	3,489	317	11x
	Central African Republic	Africa	2,958	290	10x
	Angola	Africa	1,947	327	6x
	Ghana	Africa	682	201	3.4x
	Côte d'Ivoire	Africa	646	525	1.2x
	Burkina Faso	Africa	212	173	1.2x
	Brazil	Americas	212	200	1.1x

4 How much does the reporting rate vary over time?

Table 5: Within-country variability of the implied reporting rate, 1990-2023. Countries with at least four years with a rate.

Cause	Countries	Median rate	Median fold range	Range >= 10x	Within-country share	Rate rising
Diphtheria	69	14.4%	42x	88%	51%	61%
Pertussis	175	0.6%	134x	92%	36%	43%
Tetanus	109	18.9%	20x	64%	38%	61%

Table 5 and Figure 7 summarise the spread. Pertussis reporting is the least stable: 92% of countries have a rate spanning at least a factor of ten. Tetanus is the most stable, but 64% of countries still span tenfold. Pertussis rates are falling in most countries (57%). Diphtheria and tetanus rates are rising in most.

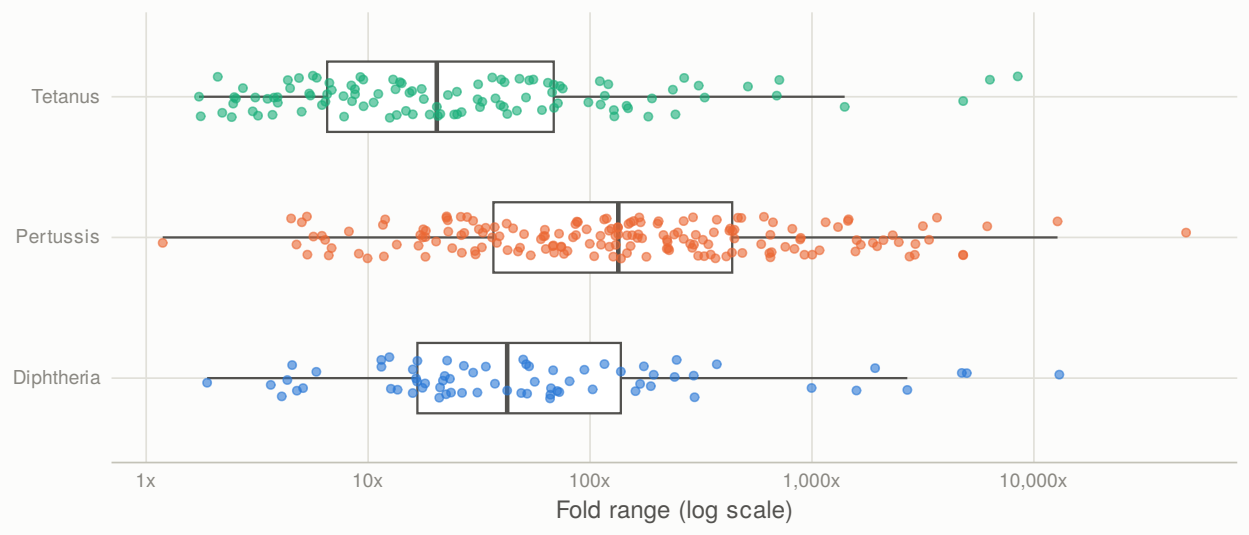


Figure 7: Fold range of the implied reporting rate within each country (highest ÷ lowest non-zero rate, 1990-2023). Each point is a country.

4.1 Country-by-year heatmaps

The heatmaps show every country with at least five evaluable years. Darker cells mean a higher reported share of modelled cases. Within each WHO region, countries are sorted by median rate, highest at the top. Grey cells are years with zero reported cases. Blank cells are years with no report to WHO, or with GBD incidence below 10 cases.

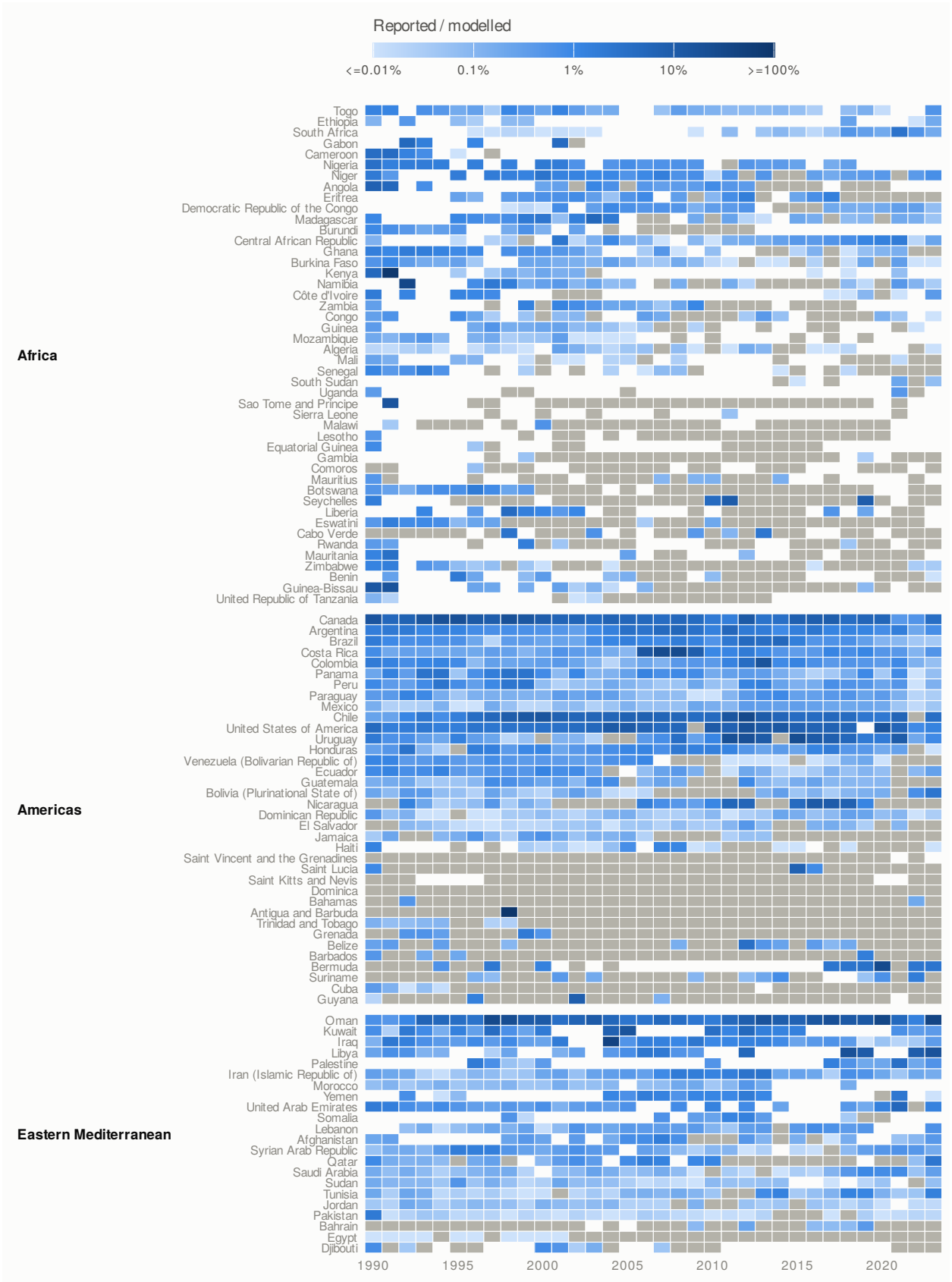


Figure 8: Pertussis: implied reporting rate by country and year (Africa, Americas, Eastern Mediterranean).

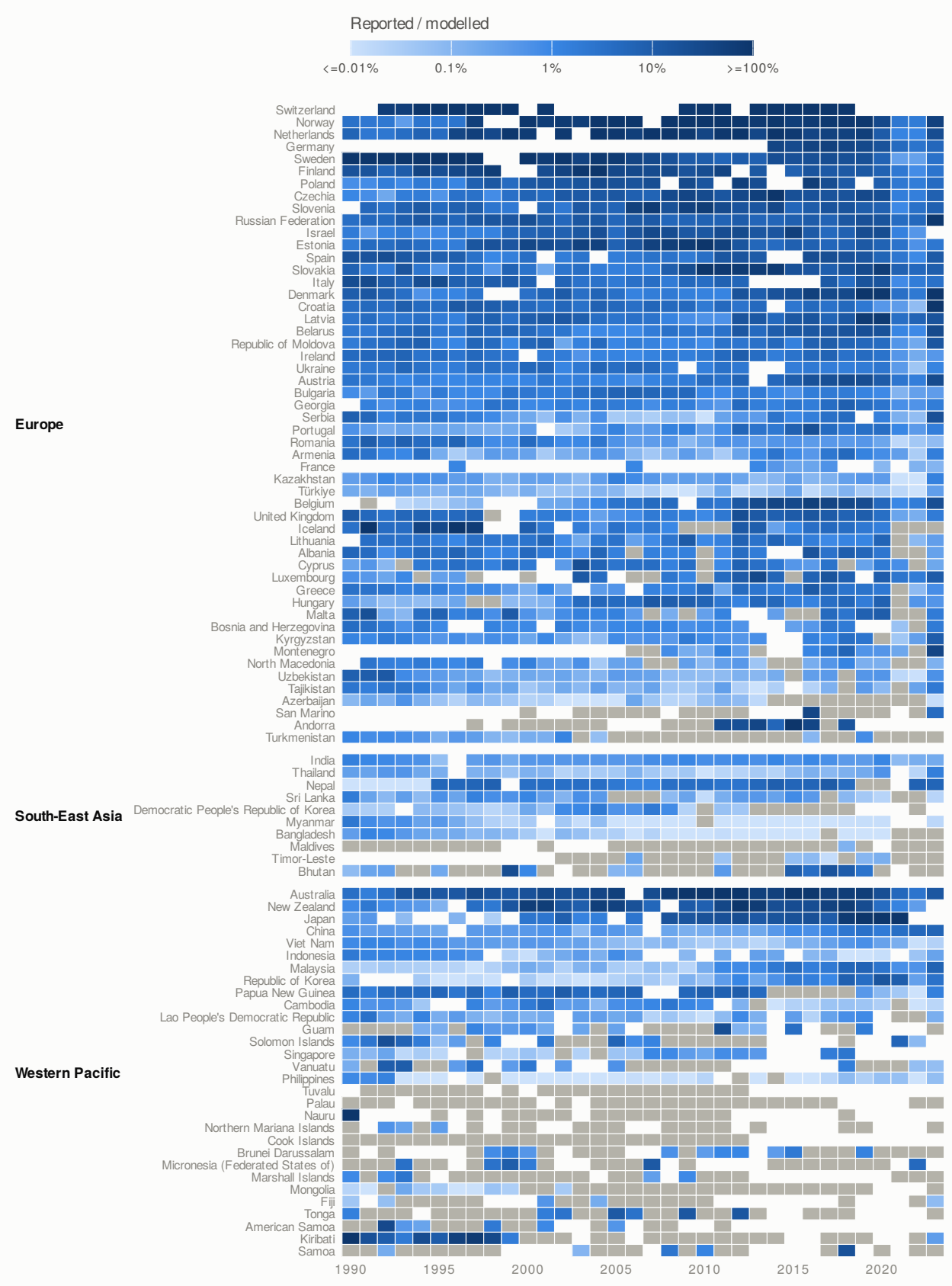


Figure 9: Pertussis: implied reporting rate by country and year (Europe, South-East Asia, Western Pacific).

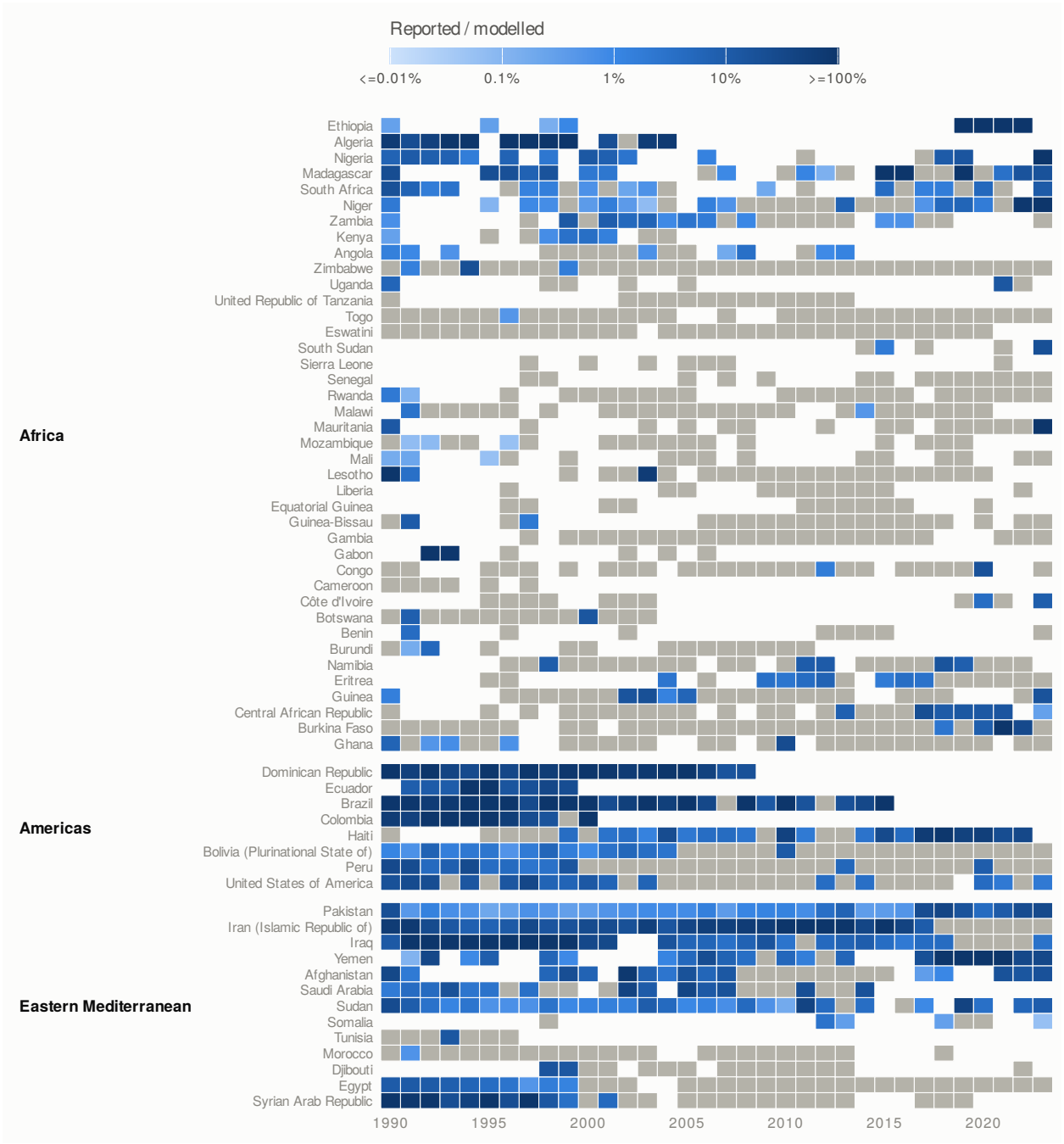


Figure 10: Diphtheria: implied reporting rate by country and year. Few countries have five or more years with at least 10 modelled cases.

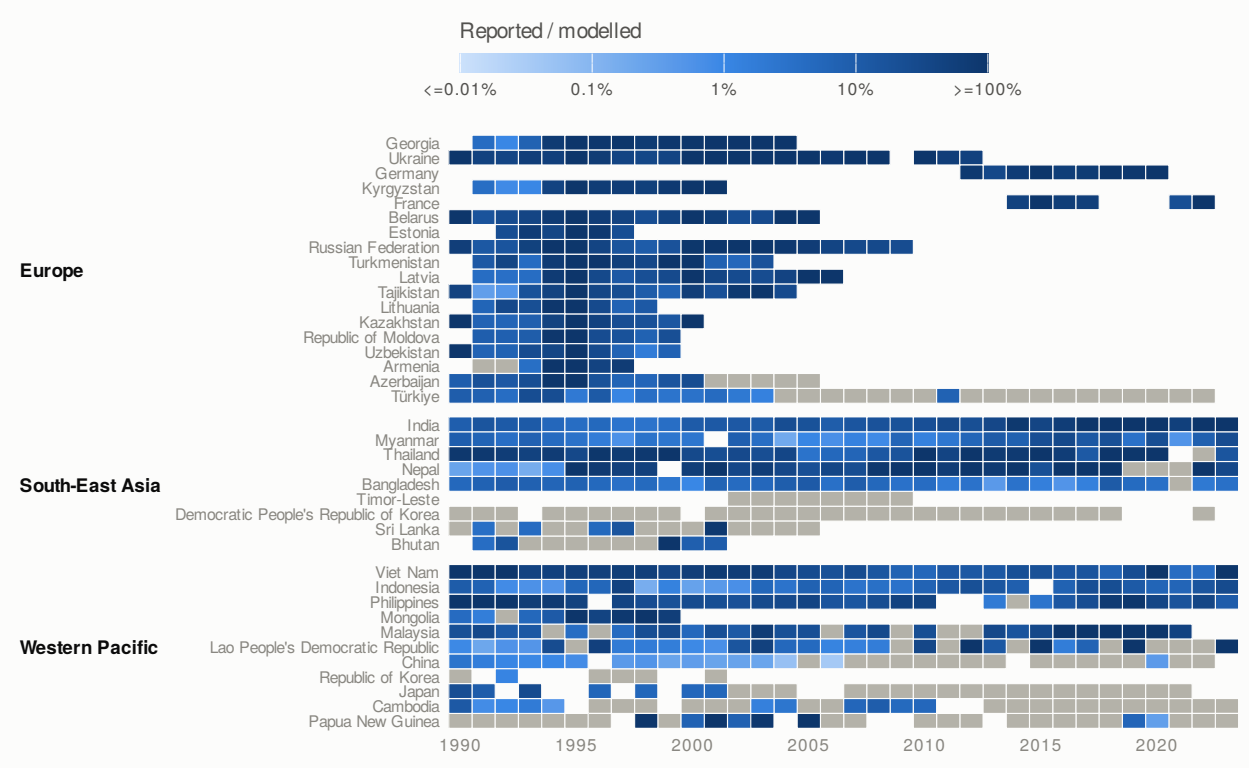


Figure 11: Diphtheria: implied reporting rate by country and year (Europe, South-East Asia, Western Pacific).

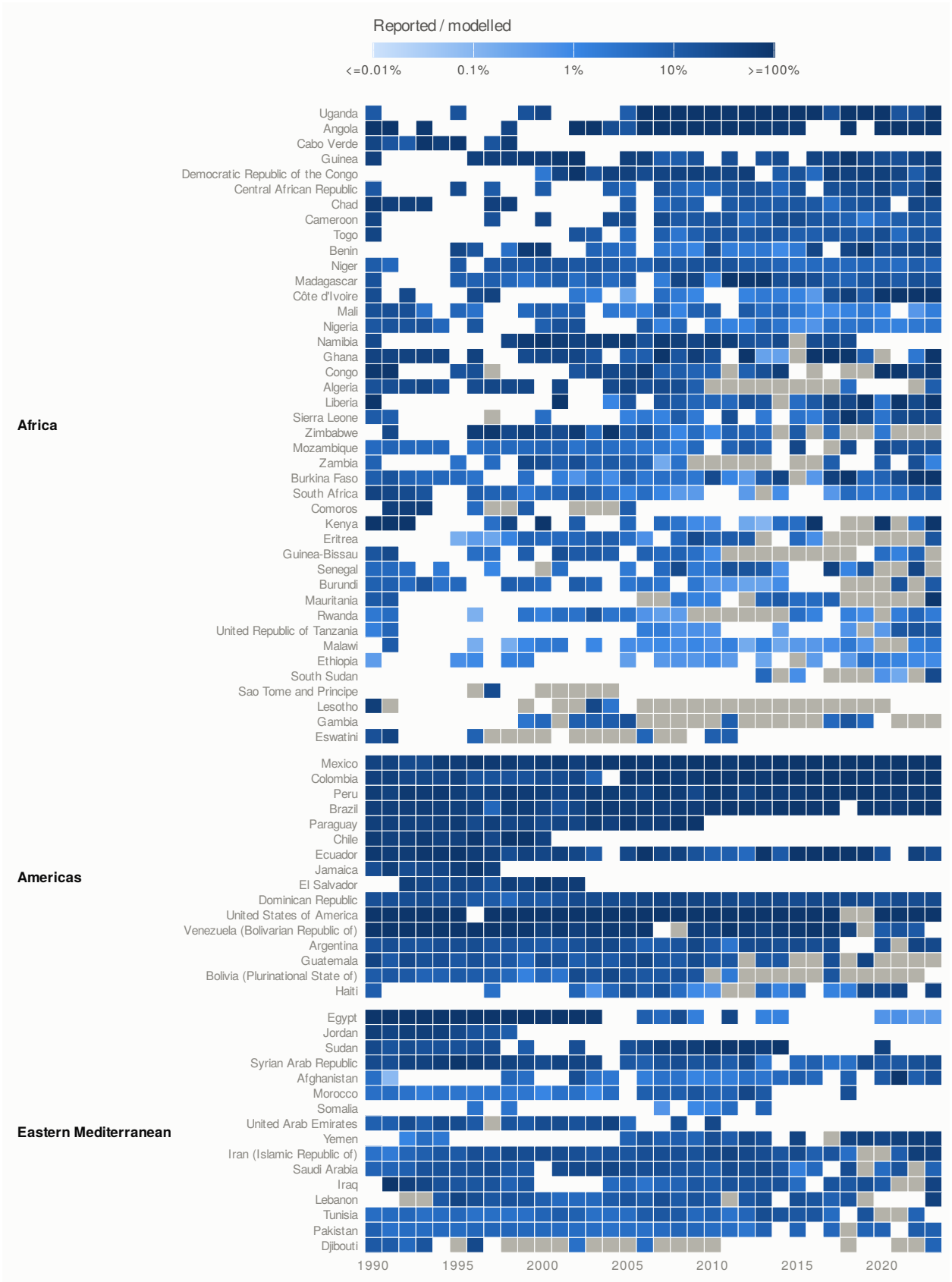


Figure 12: Tetanus: implied reporting rate by country and year (Africa, Americas, Eastern Mediterranean).

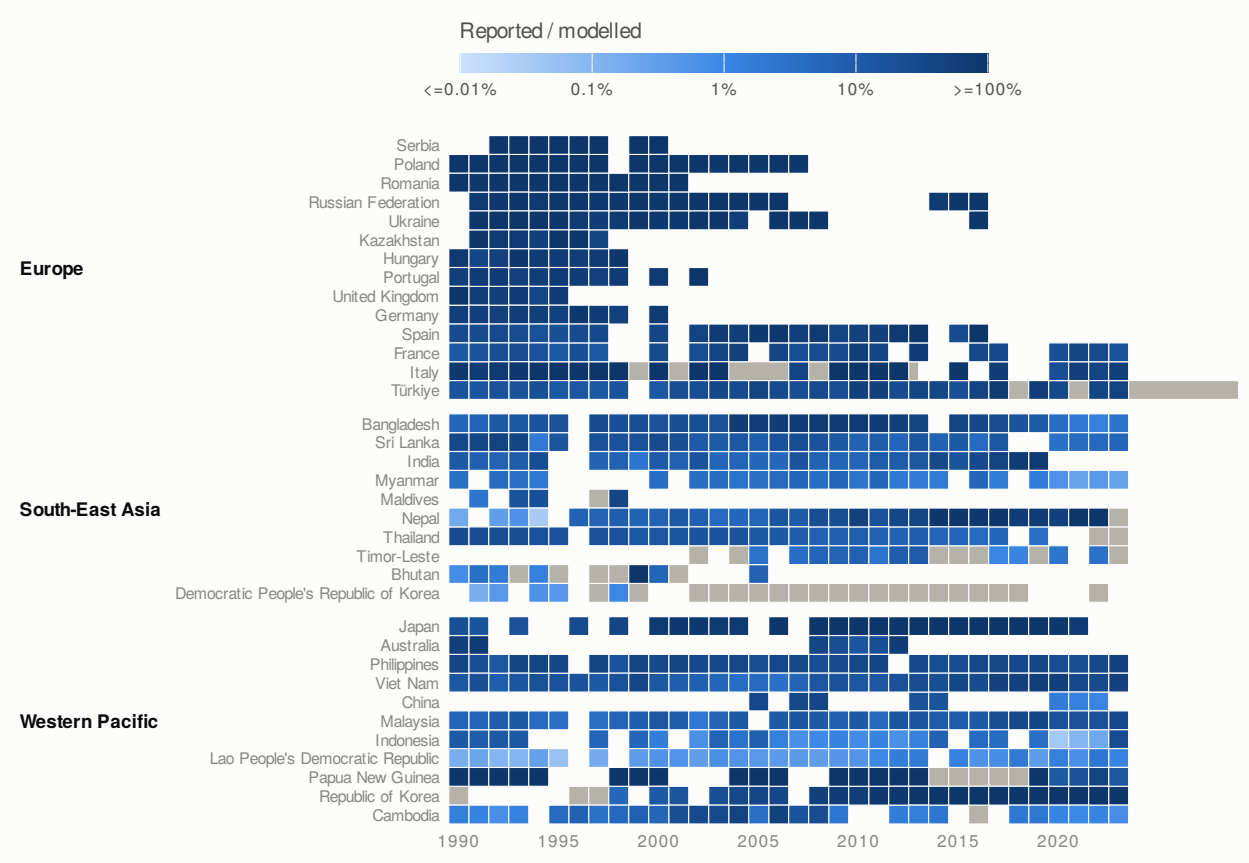


Figure 13: Tetanus: implied reporting rate by country and year (Europe, South-East Asia, Western Pacific).

4.2 High-burden countries

Figure 14, Figure 15 and Figure 16 follow the reporting rate in the 12 countries with the most modelled cases in 2023. Even in large countries with established surveillance, the rate can shift by one or two orders of magnitude within a few years.

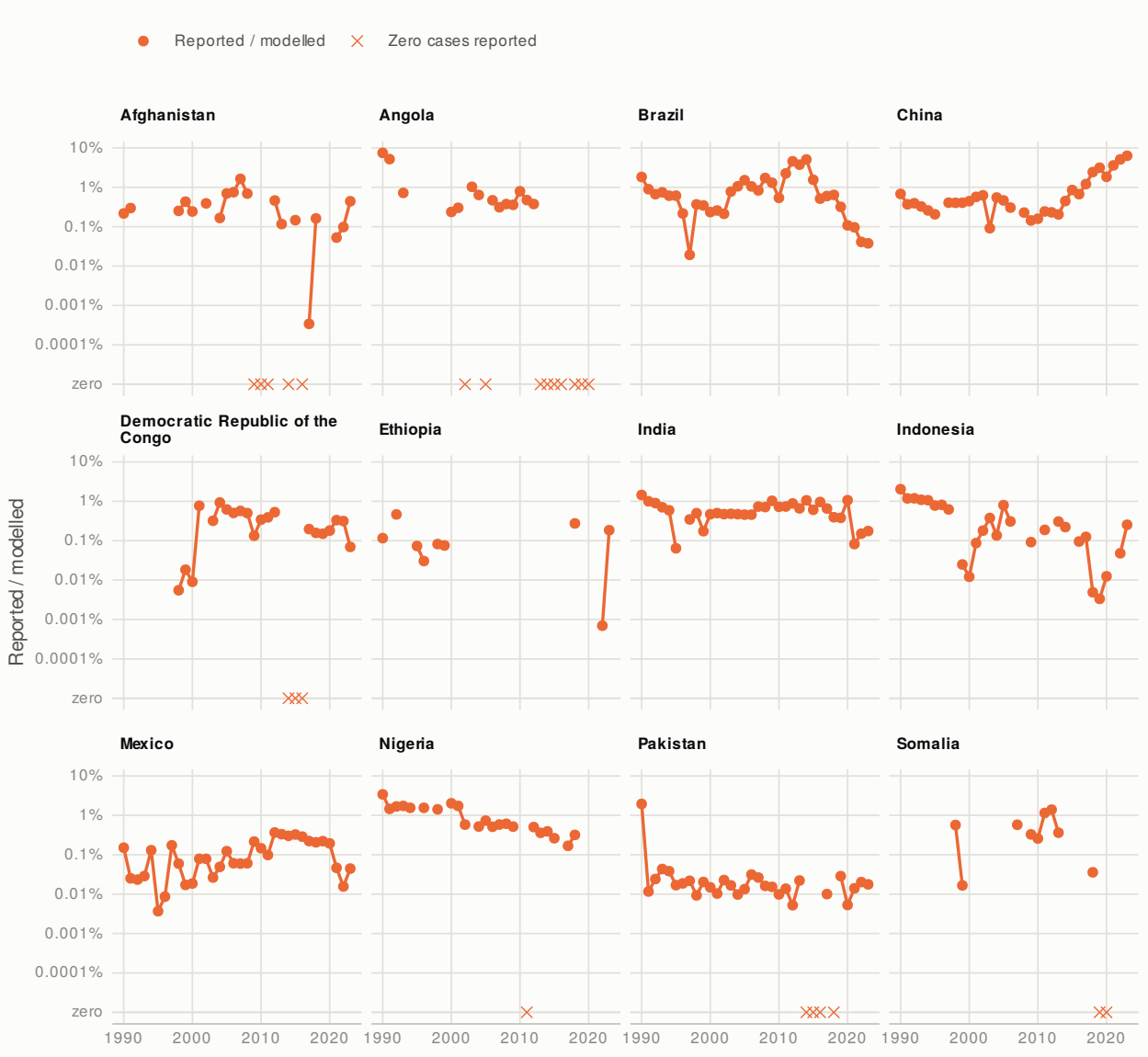


Figure 14: Pertussis: implied reporting rate over time in the 12 highest-burden countries. Log scale; × marks years with zero reported cases; gaps are years with no rate.

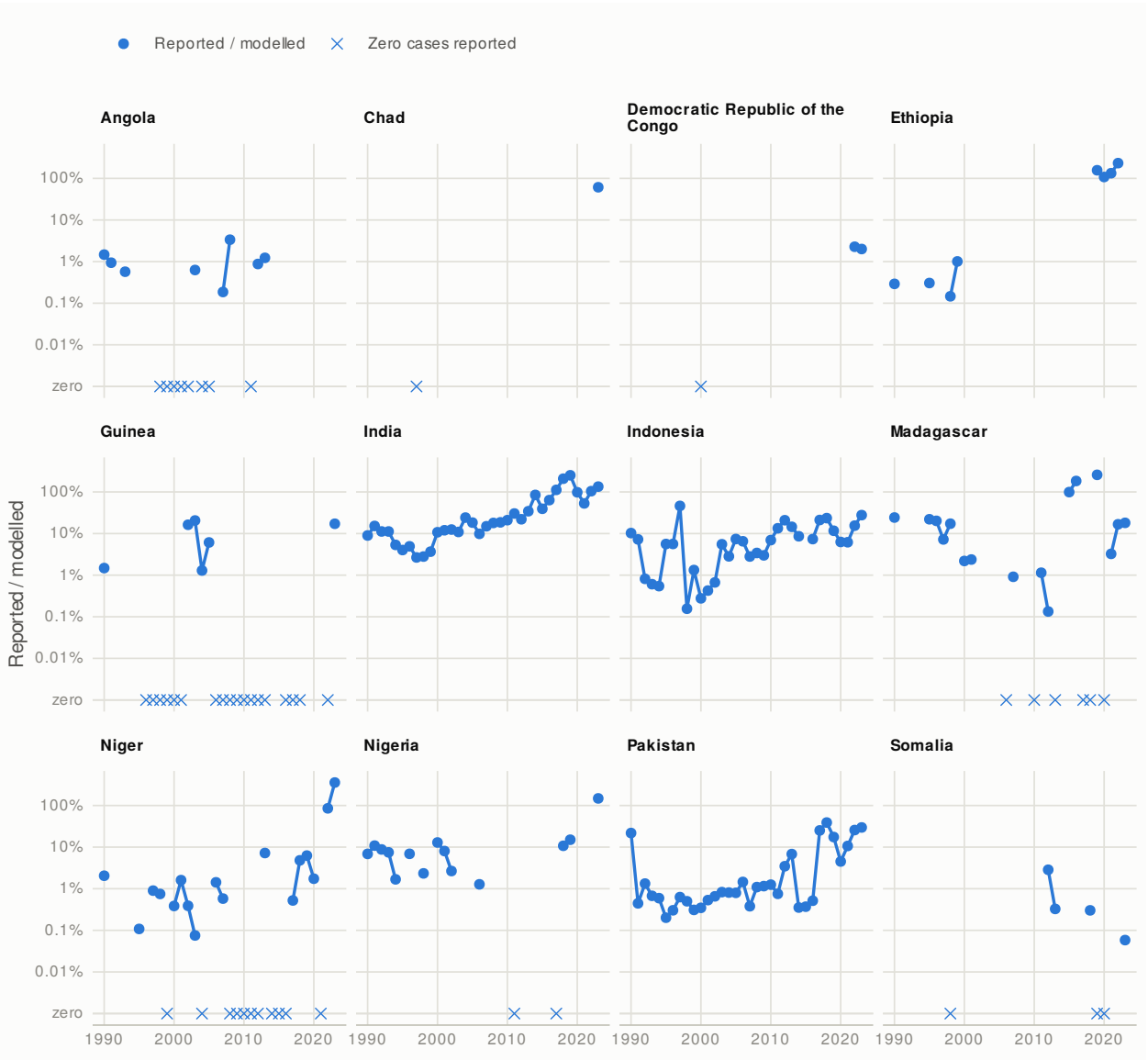


Figure 15: Diphtheria: implied reporting rate over time in the 12 highest-burden countries.

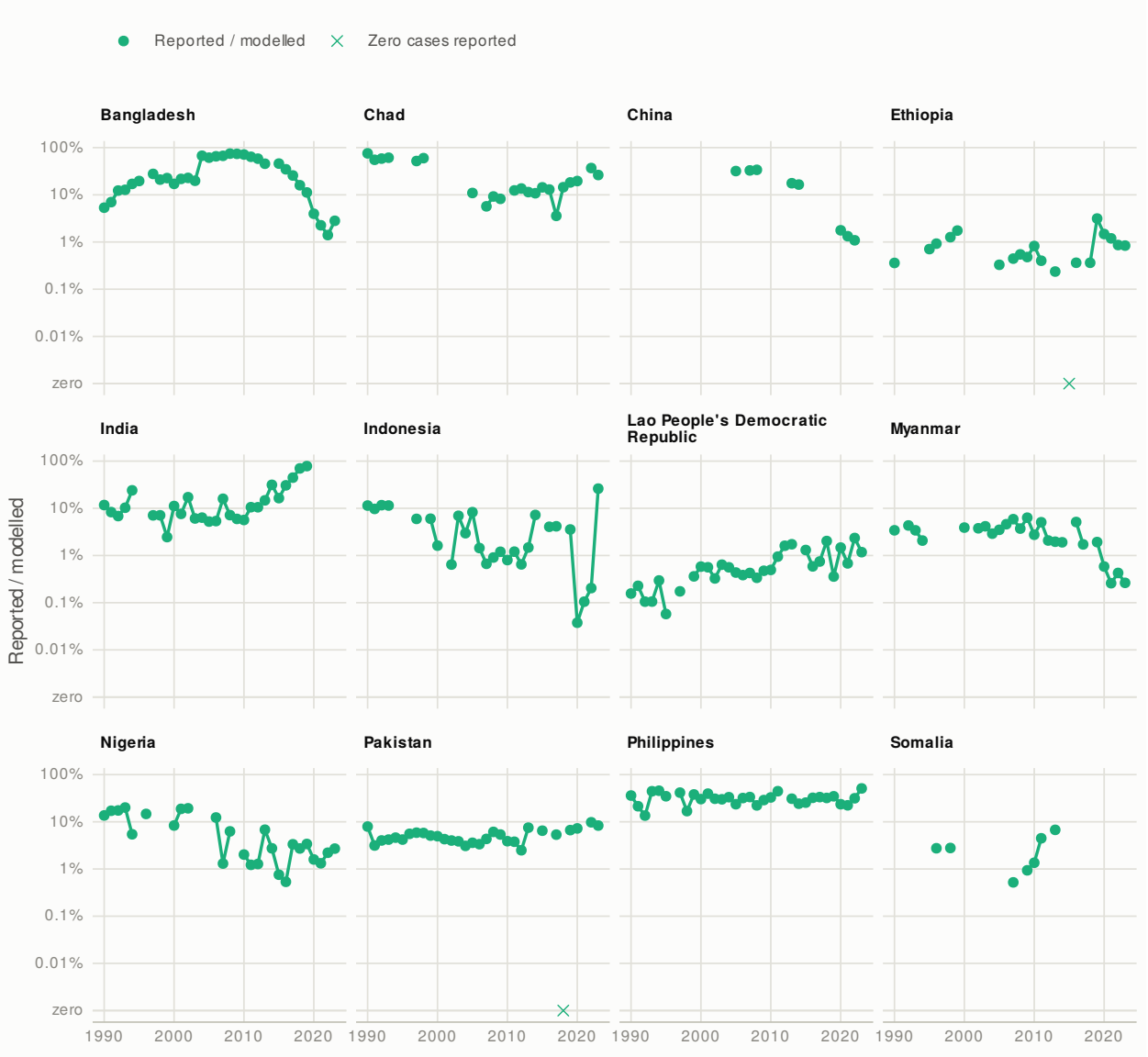


Figure 16: Tetanus: implied reporting rate over time in the 12 highest-burden countries.

5 Where do the sources disagree most?

Pooling 2020-2023 smooths single-year outbreaks. Figure 17 shows the ten countries at each end per cause. At the high end, diphtheria and tetanus reports exceed GBD's estimate several-fold in outbreak and conflict settings. At the low end, 47 countries sent WHO only zeros for pertussis in the years they reported, despite GBD estimating at least 10 cases a year in each. Only 18 of them reported zero in all four years; the rest have years with no report at all.

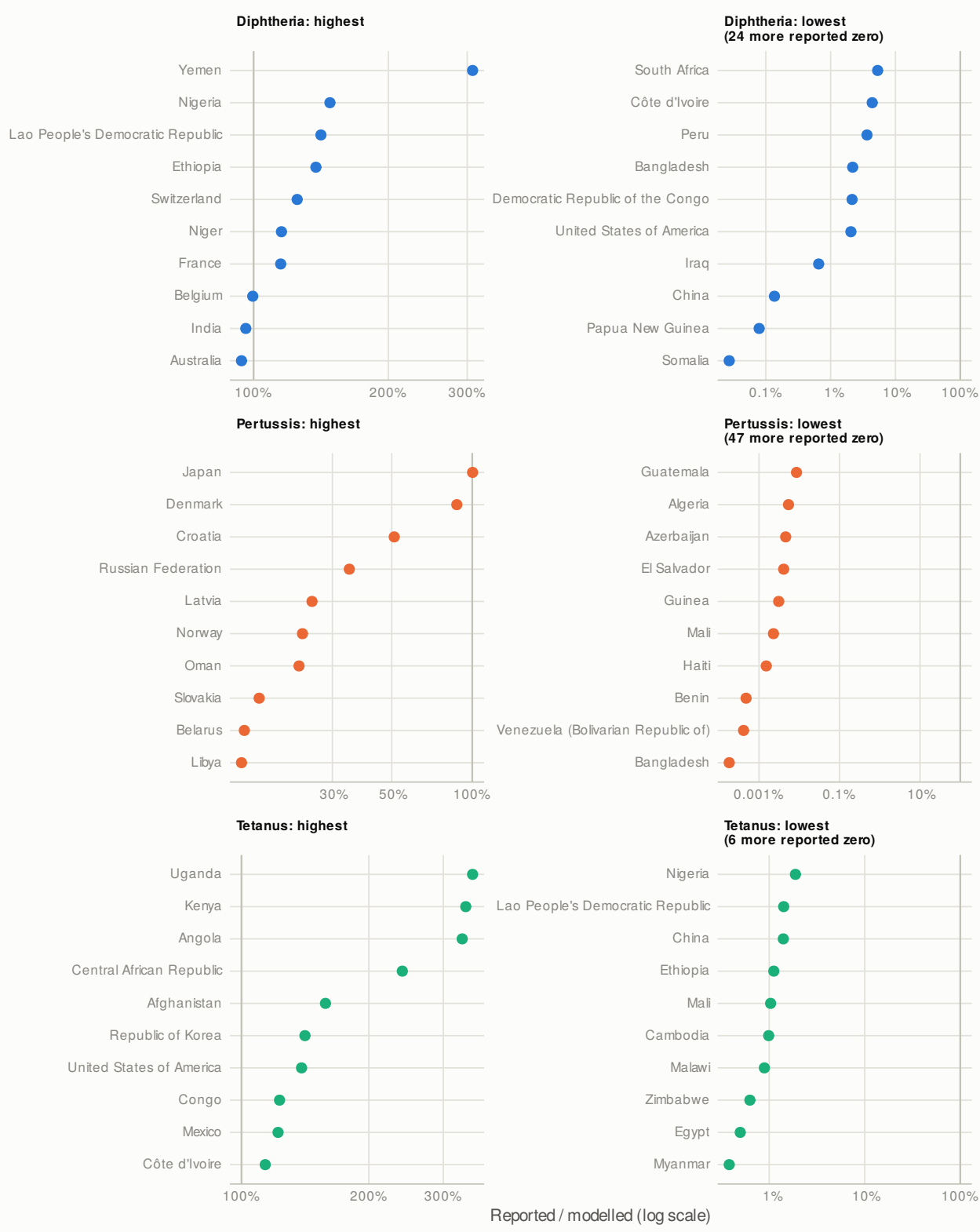


Figure 17: Countries with the highest and lowest reported ÷ modelled cases, 2020-2023 pooled. The vertical line marks reported = modelled. Countries reporting zero throughout are counted in the panel titles.

6 Is the reporting rate related to vaccine coverage?

Table 6: Association between DTP3 coverage and log reporting rate. *Between*: country means. *Within*: deviations from each country's own mean (countries with at least four years).

Cause	Country-years	Countries	Spearman, pooled	Spearman, between	Pearson, within
Diphtheria	606	75	0.36	0.42	0.23
Pertussis	2,636	176	0.34	0.38	0.03
Tetanus	1,573	102	0.22	0.32	0.05

Higher DTP3 coverage goes with a higher reporting rate across countries (Table 6, Figure 18). For pertussis, the rate rises steeply once coverage passes about 85%. Within countries, the association is much weaker (Pearson 0.03 for pertussis, 0.05 for tetanus and 0.23 for diphtheria). Coverage mostly marks which countries have stronger health systems and surveillance; it says little about why one country's rate changes over time. Part of this relationship is mechanical, because GBD uses vaccine coverage as a covariate in its incidence models.

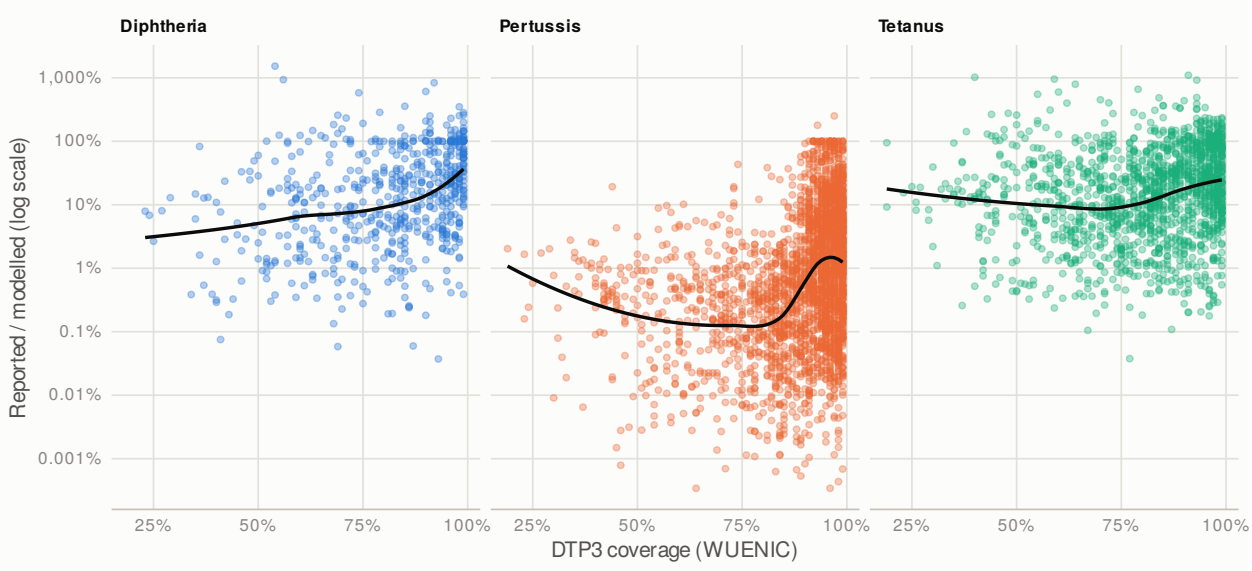


Figure 18: Implied reporting rate vs DTP3 coverage, all country-years 1990-2023. Black line: loess smooth. Excludes zero-report years and GBD incidence below 10 cases.

7 Time trends

GBD shows a smooth decline in all three diseases (Figure 19). Reported cases are far more volatile: diphtheria reports have risen since the mid-2010s, and pertussis reports reached 941,900 in 2024. GBD's pertussis estimate dips in 2020-2021 and recovers in 2022-2023. Reported totals also depend on how many countries report each year.

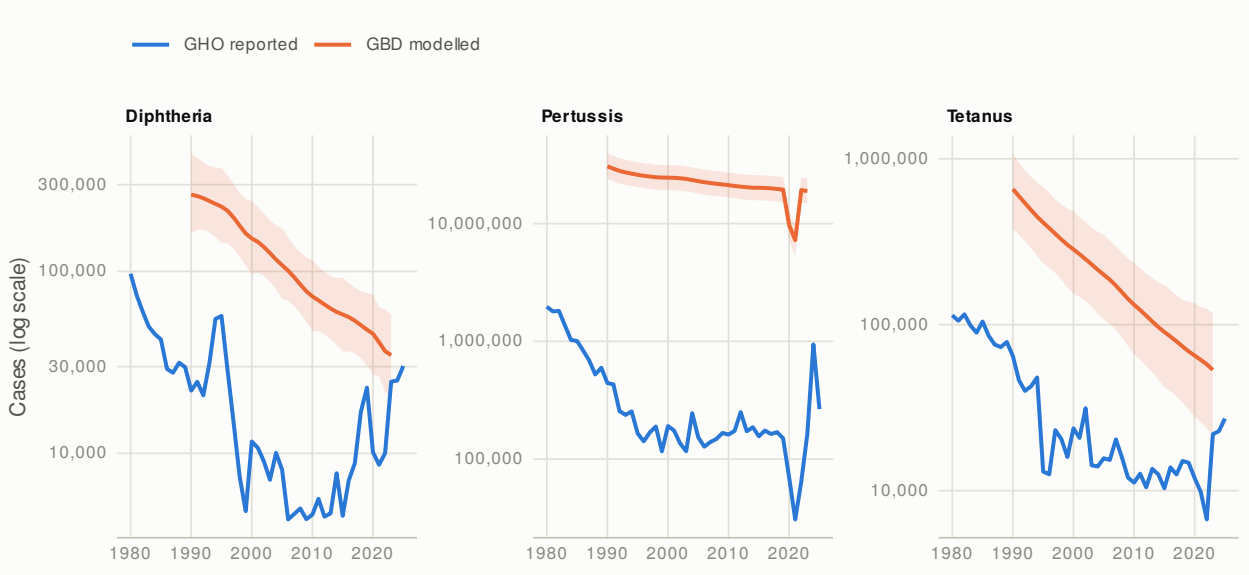


Figure 19: Global cases, WHO reported vs GBD modelled incidence, summed over countries in both sources. Log scale. Band: sum of country 95% UIs, which overstates the aggregate interval.

7.1 Which countries make up the global totals

The two global totals in Figure 19 are driven by different countries (Figure 20, Figure 21, Figure 22, Table 7).

- **GBD totals follow population and vaccine coverage.** In 2023 the largest GBD contributors are India (15%) and Nigeria (11%) for pertussis, Nigeria (28%) and Indonesia (10%) for diphtheria and Indonesia (18%) and India (10%) for tetanus.
- **WHO totals follow whoever is reporting an outbreak.** A handful of countries make up most of each year's reported total, and they change from year to year. Diphtheria moved from India (2019) to Nigeria (2023-2025). In 2024, China alone accounted for half of the global pertussis count, after it broadened its case definition ("Epidemiological and Spatial–Temporal Clustering Characteristics of Pertussis in Southwest China, 2013–2024" 2025).
- **Historic peaks were concentrated too.** The 1993-1995 diphtheria peak in WHO data comes from the epidemic in the former Soviet Union. In the early 1980s China and India were the largest single contributors to reported pertussis.

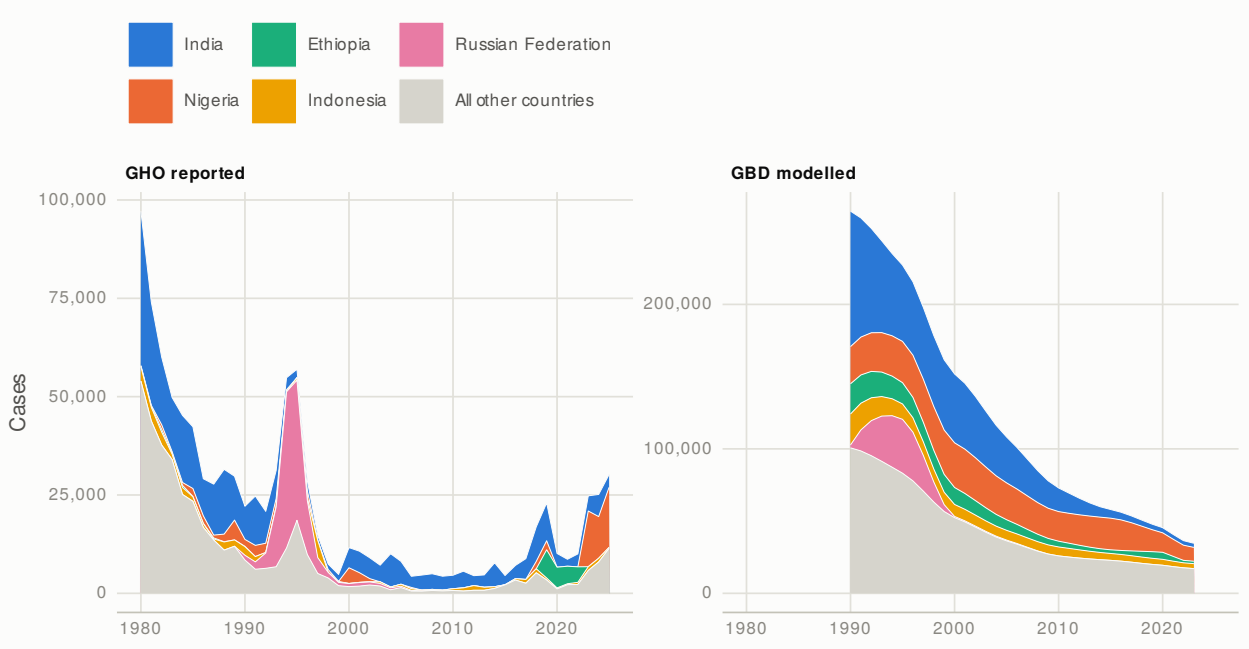


Figure 20: Diphtheria: contribution of the largest countries to the global WHO-reported and GBD-modelled totals. Countries shown are the union of each source's top four since 1990.

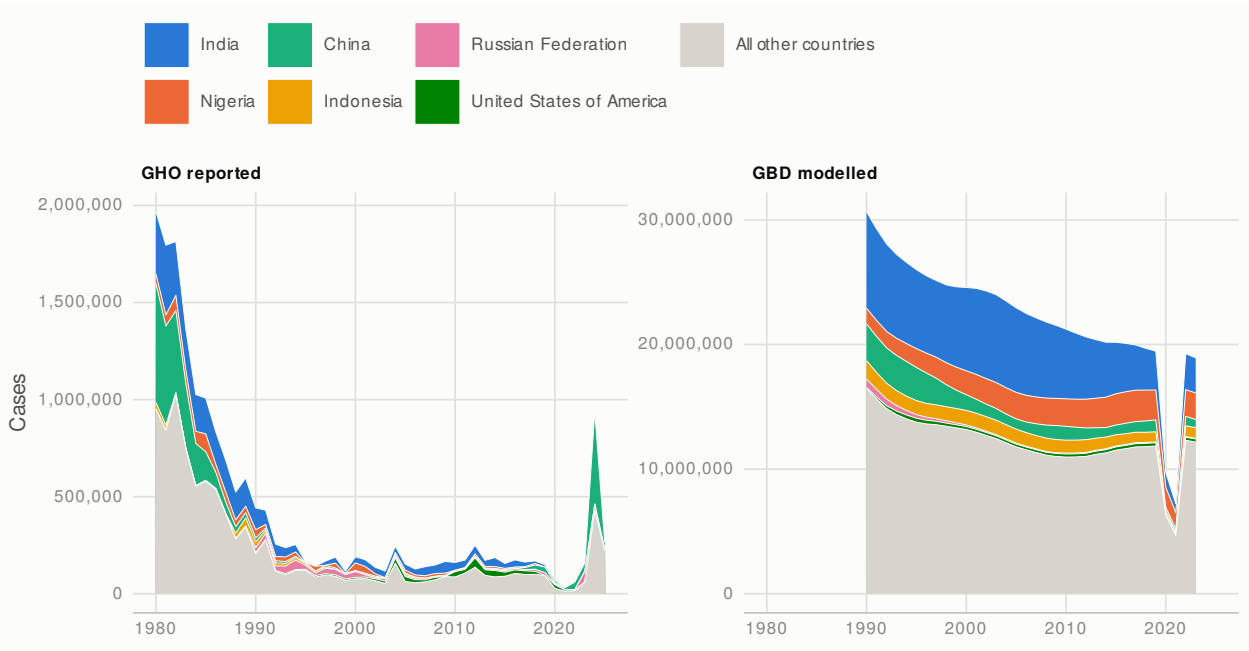


Figure 21: Pertussis: contribution of the largest countries to the global totals.

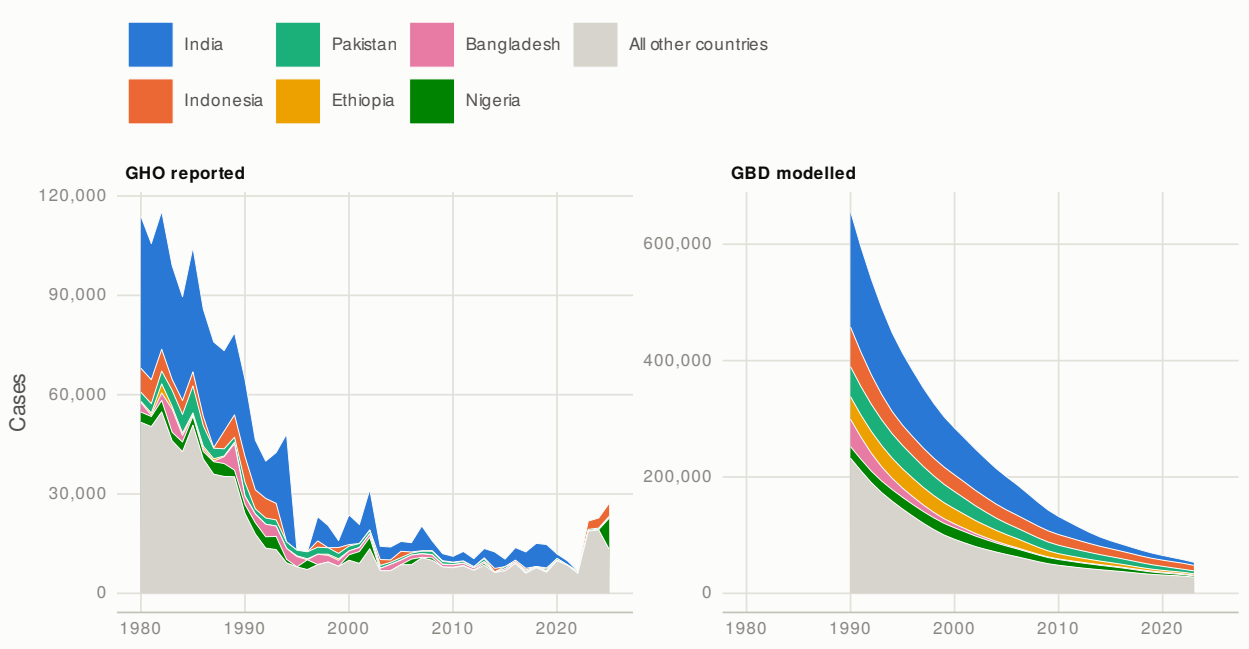


Figure 22: Tetanus: contribution of the largest countries to the global totals.

Table 7: Largest contributors to the WHO-reported global total in selected years (cases and share of the global total).

Disease	Year	Global total	Top three countries	Top-three share
Diphtheria	2010	4,603	India (3,434, 75%); Indonesia (432, 9%); Nepal (146, 3%)	87%
	2019	22,989	India (9,622, 42%); Ethiopia (7,184, 31%); Nigeria (2,289, 10%)	83%
	2023	24,782	Nigeria (14,163, 57%); India (3,850, 16%); Niger (3,286, 13%)	86%
	2024	25,147	Nigeria (10,595, 42%); India (5,634, 22%); Chad (2,599, 10%)	75%
	2025	30,206	Nigeria (15,273, 51%); Somalia (3,622, 12%); India (3,249, 11%)	73%
Pertussis	2010	161,279	India (40,508, 25%); Australia (34,285, 21%); United States of America (27,550, 17%)	63%
	2019	149,883	China (30,027, 20%); Japan (16,845, 11%); Russian Federation (14,407, 10%)	41%
	2023	163,466	Russian Federation (52,783, 32%); China (41,124, 25%); Denmark (6,059, 4%)	61%
	2024	941,900	China (476,690, 51%); Australia (57,167, 6%); Republic of Korea (48,454, 5%)	62%
	2025	265,337	Japan (89,577, 34%); China (33,364, 13%); Australia (25,277, 10%)	56%
Tetanus	2010	11,181	Uganda (1,822, 16%); India (1,756, 16%); Philippines (1,140, 10%)	42%
	2019	14,745	India (7,071, 48%); Uganda (1,360, 9%); Philippines (953, 6%)	64%
	2023	21,891	Kenya (4,463, 20%); Uganda (3,489, 16%); Central African Republic (2,958, 14%)	50%
	2024	22,821	Kenya (4,306, 19%); Central African Republic (3,440, 15%); Uganda (3,094, 14%)	48%
	2025	27,200	Nigeria (9,331, 34%); Kenya (3,992, 15%); Indonesia (3,919, 14%)	63%

At country level (Figure 23, Table 8), reported and modelled cases move in the same direction from one year to the next in 53% of country-years for pertussis, which is little better than chance. The figure is 53% for tetanus and 62% for diphtheria.

Table 8: Agreement between year-on-year country-level changes in reported and modelled cases.

Cause	Country-years	Same direction	Correlation of log changes
Diphtheria	865	62%	0.20
Pertussis	3,290	53%	0.16
Tetanus	1,851	53%	0.02

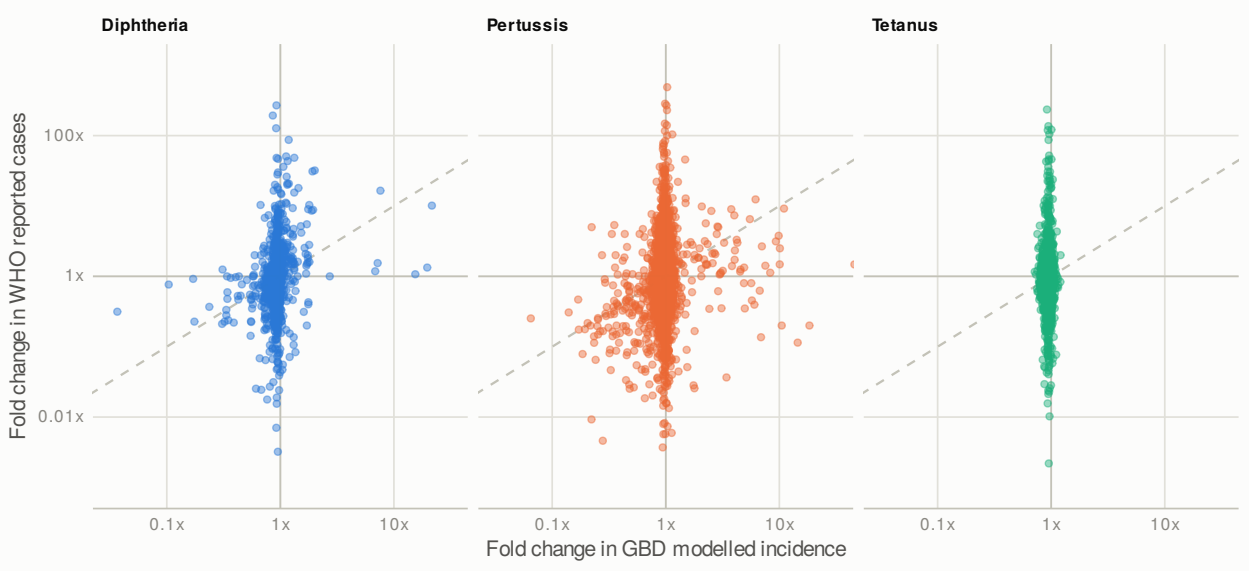


Figure 23: Year-on-year fold change in reported vs modelled cases, one point per country-year. Dashed line: equal change.

8 What explains these patterns?

A research brief compiled from official documentation and the literature (VIMC GBD–WHO comparison project 2026) checked each pattern in this report against published sources. Most of the discrepancies follow from documented GBD 2023 modelling choices, and two are artefacts of the WHO data. Table 9 summarises the findings; the full brief, with quotations and page references, is in `docs/research_brief.md`.

Table 9: Patterns in this report and what the documentation says about them. Status: Explained = a source states it; Plausible = a source describes a mechanism that fits; Artefact = a feature of the WHO data.

Pattern	Explanation	Status	Sources
Pertussis reports about 0.6% of GBD	GBD fits a regression to WHO notifications, then gives every country Switzerland's random effect, assuming Switzerland reports completely. Switzerland's own ratio in our data is 100%.	Explained	GBD 2023 nonfatal appendix; GBD pertussis 2026
Diphtheria ratios of exactly 1	New in GBD 2023: estimates below the WHO count are replaced by the count (also for pertussis, not tetanus).	Explained	GBD 2023 nonfatal appendix
Diphtheria and tetanus above GBD where GBD is near zero	Incidence = modelled deaths / case-fatality ratio, so few deaths give near-zero incidence. Tetanus has no floor. European diphtheria cases are largely cutaneous or imported; GBD counts respiratory diphtheria.	Plausible	GBD 2023 appendices; Brockhaus 2024
Tetanus above GBD in Uganda, Kenya, Angola, CAR	Death-derived incidence with high case-fatality ratios in low-HAQ settings; no country-specific documentation found.	Plausible	GBD 2023 nonfatal appendix
Year-to-year changes do not line up	GBD estimates vary only with smoothed covariates (5-year DTP3, masking) and smoothed deaths; GBD reports country concordance with WHO of r = -0.10 for pertussis in 2023.	Partly documented	GBD pertussis 2026; Li 2026
Reporting rate higher where DTP3 is higher	Partly built in: DTP3 is a covariate in all three GBD models. Surveillance sensitivity also tracks income.	Plausible	Li 2026; Chongqing 2025
GBD pertussis dip 2020-2021	Driven by a mask-use covariate added in GBD 2023, not by notifications.	Explained	GBD 2023 nonfatal appendix
WHO pertussis surge 2024	Post-COVID-19 immunity gap and wider PCR testing; China's new case criteria from 2024.	Explained	ECDC 2024; Chongqing 2025
China pertussis reporting rate rising	PCR confirmation accepted from 2017; broader criteria from 2024; also genuine resurgence and macrolide resistance.	Explained	Chongqing 2025; Dai 2024
India tetanus collapse after 2019	From 2022 India's total tetanus equals its neonatal count; non-neonatal cases are no longer reported. Excluded here.	Artefact	GHO API
Persistent zero reporters	Many are blanks (no report), not zeros. Pertussis surveillance may be sentinel-only, so a zero is not evidence of absence.	Artefact / plausible	GHO API; WHO 2018 standards

Three implications follow for how the reporting rate should be read.

- **Pertussis:** the absolute level rests on the Switzerland assumption. GBD's own sensitivity analysis shows alternative reference choices would change the scaling by a factor of 0.39 to 0.0069 (GBD 2023 Pertussis Collaborators 2026). Comparisons over time within a country are more robust than levels between countries.
- **Diphtheria:** ratios near 1 are copies, not agreement, and ratios above 1 in GBD 2023 point to WHO figures added or revised after GBD's data freeze, or to case definitions that differ (cutaneous diphtheria).
- **Tetanus:** the ratio compares notifications with a death-based estimate. A ratio above 1 is not over-reporting in itself; in high-income settings the notification is probably the better count.

The brief also found that this is not a new question for pertussis: two 2026 papers compare GBD and WHO pertussis data directly (GBD 2023 Pertussis Collaborators 2026; Li et al. 2026). No published reconciliation was found for diphtheria or tetanus, and the 50-year EPI impact analysis used GBD 2021 burden for DTP without reconciling it with WHO cases (Shattock et al. 2024).

9 Limitations

- **The sources are not independent.** GBD uses WHO notifications as its main input for pertussis and as a floor for diphtheria (Section 8). Close agreement partly reflects that rather than independent confirmation.
- **Reports can exceed GBD's estimates.** This happens mainly where GBD's death-derived estimate is near zero, and for tetanus in parts of Africa. In those settings the ratio is not a completeness measure.
- **The threshold excludes many country-years.** The 10-case threshold removes most diphtheria country-years in low-incidence settings. Variability results for diphtheria rest on 69 countries.
- **More years widen the range.** The fold range is the gap between a country's most extreme years, so it grows with the number of years observed. Compare it across causes, not with summaries based on fewer years.
- **GBD estimates are smoothed over time.** Year-to-year movement in the ratio therefore mostly reflects reported cases, not the model.
- **Only all-age, both-sex totals are compared.** GHO case counts are not disaggregated, so age patterns cannot be compared. A cleaner tetanus comparison would set WHO neonatal tetanus against GBD tetanus in neonates, which needs an age-specific GBD export.
- **Floored and pandemic years are kept in the rate analysis.** Floored diphtheria and pertussis country-years, and pertussis in 2020-2022, are labelled but not excluded. Excluding them is a natural sensitivity analysis.

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VIMC GBD–WHO comparison project. 2026. *DTP Data: GBD 2023 Vs WHO Reported Cases – Research Brief*. `docs/research_brief.md` in this repository.

10 Appendix

Table 10: Countries and territories present in only one source.

Source	ISO3	Country
GBD only	GRL	Greenland
GBD only	PRI	Puerto Rico
GBD only	TWN	Taiwan
GBD only	VIR	U.S. Virgin Islands
GHO only	AIA	Anguilla
GHO only	ABW	Aruba
GHO only	VGB	British Virgin Islands
GHO only	CYM	Cayman Islands
GHO only	CUW	Curaçao
GHO only	PYF	French Polynesia
GHO only	HKG	Hong Kong SAR China
GHO only	XKX	Kosovo
GHO only	MAC	Macao SAR China
GHO only	MSR	Montserrat
GHO only	NCL	New Caledonia
GHO only	SXM	Sint Maarten
GHO only	TCA	Turks & Caicos Islands
GHO only	WLF	Wallis & Futuna

Data: Global Burden of Disease Collaborative Network. *Global Burden of Disease Study 2023 (GBD 2023) Results.* Seattle: IHME, 2024. World Health Organization, Global Health Observatory (accessed 22 October 2026).